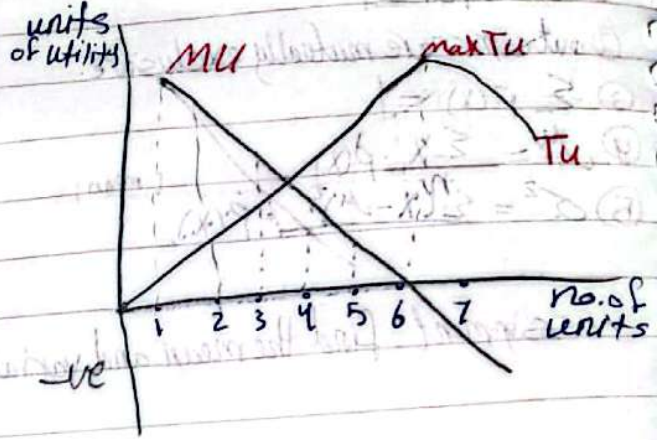
Cardinal utility.

X	Total utility of x	marginal utility of x
1	15	15
2	20	10
3	30	5

(increasing at decreasing rate)



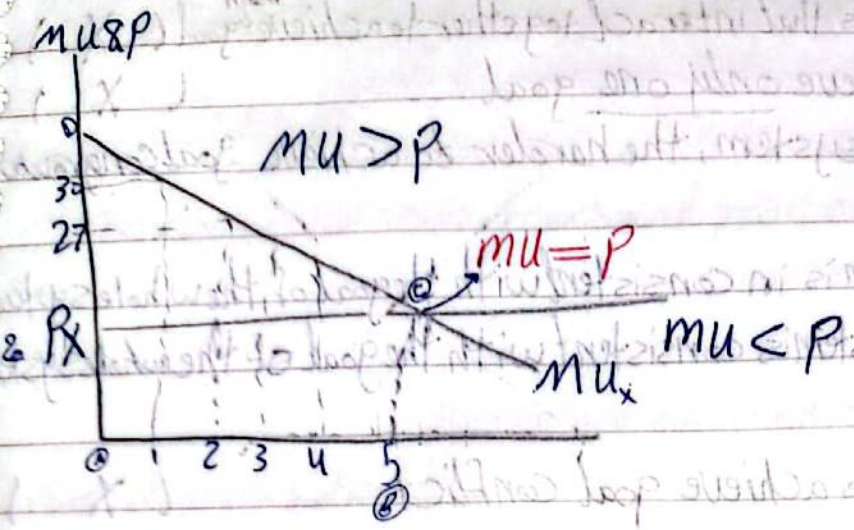
assumptions:

1. measurability.
2. Constancy. → P, income
3. Rationality. (consumer aims to max U)
4. utility is independent.
5. Continuity
6. homogeinity (quality constant)
7. MU of money is constant

---

units x	TU	MU
1	20	20
2	37	17
3	51	14
4	62	11
5	68	6
6	68	0
7	64	-4
8	50	-14

* max utility ⟹ unit 6
* MU ⟹ diminishing
* TU ⟹ increasing at diminishing rate



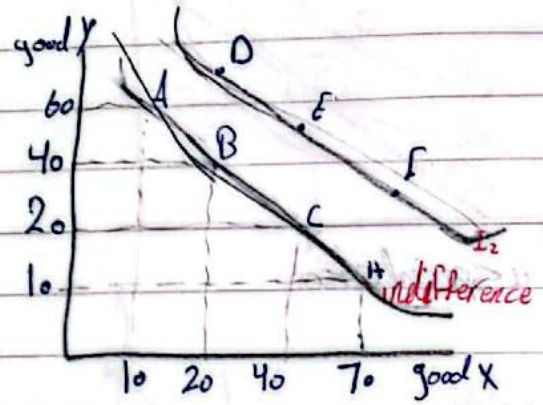
Total utility = a b C D

TP = A B C Px

Consumer Surplus = Px C D

* $\frac{MUx}{Px} = \frac{MUY}{PY} = \frac{MUz}{Pz}$ equi marginal principles.

ordinal utility

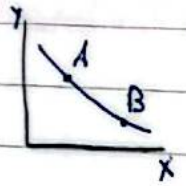


same U
60Y and 10X
40Y and 20X
20Y and 40X

* points on the same I curve give the same level of satisfaction.

Properties of indif :

① downward sloping

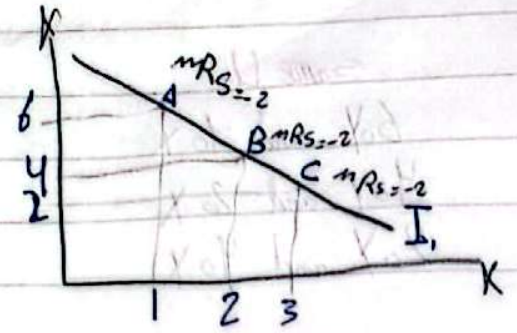


slope of Ind (MRS) $\frac{\Delta Y}{\Delta X}$

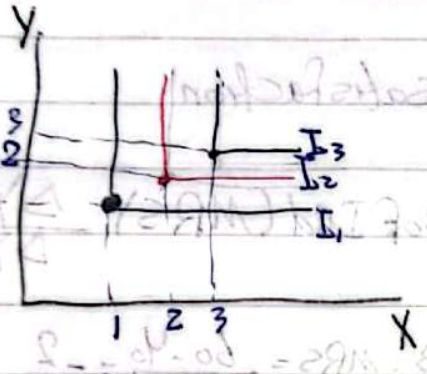
- $A \rightarrow B$: $MRS = \frac{60-40}{10-20} = -2$
- $B \rightarrow C$: $MRS = \frac{40-20}{20-40} = -1$
- $C \rightarrow H$: $MRS = \frac{20-10}{40-70} = -\frac{1}{3}$

② Indif. curves never intersect.
③ Indif. curves are convex to the origin : slope is decreasing in absolute values

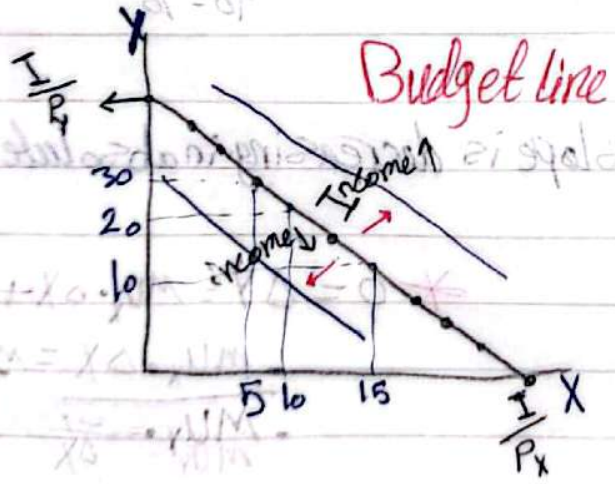
* $0 = \Delta Y = MU_x \cdot \Delta X + MU_y \cdot \Delta Y$
$-MU_x \cdot \Delta X = MU_y \cdot \Delta Y$
$-\frac{MU_x}{MU_y} = \frac{\Delta Y}{\Delta X}$



* Straight line → perfect substitutes «special case»



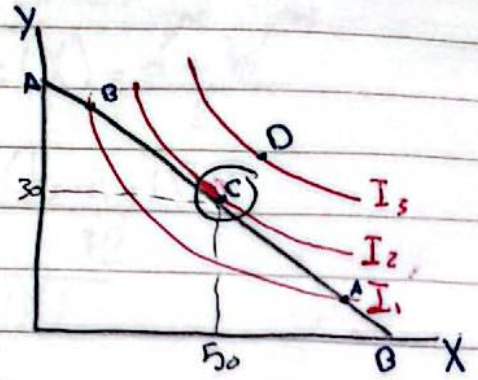
* only one point → perfect complements - «special case»



$$\frac{\Delta Y}{\Delta X} = \frac{\frac{I}{P_Y} - 0}{0 - \frac{I}{P_X}}$$

$$= -\frac{\frac{I}{P_Y}}{\frac{I}{P_X}} = \frac{I}{P_Y} \cdot \frac{P_X}{I}$$

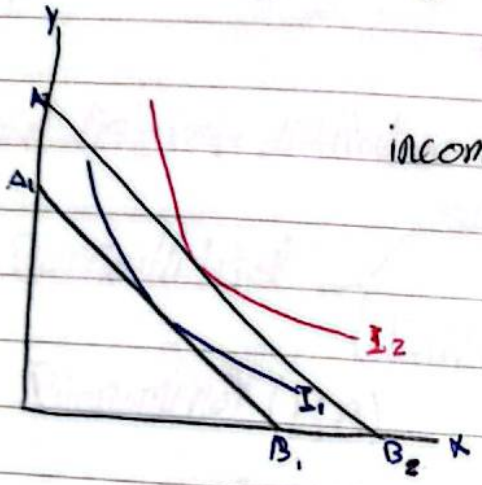
$$= -\frac{P_X}{P_Y}$$ ← Slope of Budget line.



Condition

slope of Indif. = Slope of Budget line

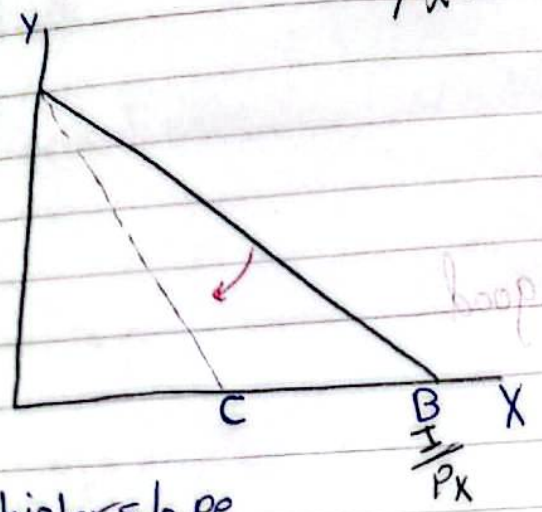
$\frac{MU_x}{MU_y} = \frac{P_x}{P_y}$



income ↑

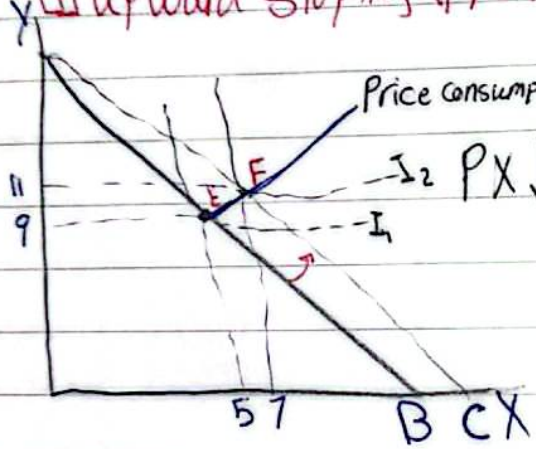
what will happen to the graph if the price changes

$P_X \uparrow$



higher slope
steeper

① upward sloping (PPC)



Price consumption curve (PCC) ⟹ ΔP_x and Q_x Q_y

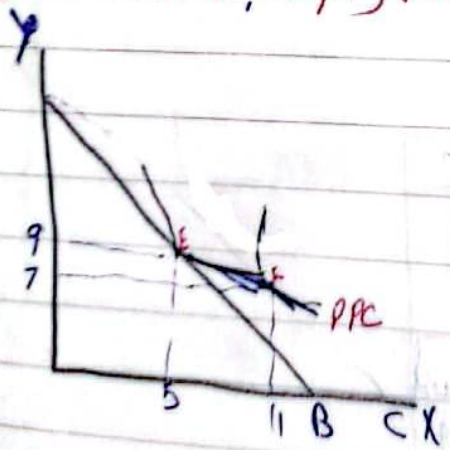
$P_X \downarrow$

$* \; \varepsilon_D = \frac{\% \Delta Q_D}{\% \Delta P}$

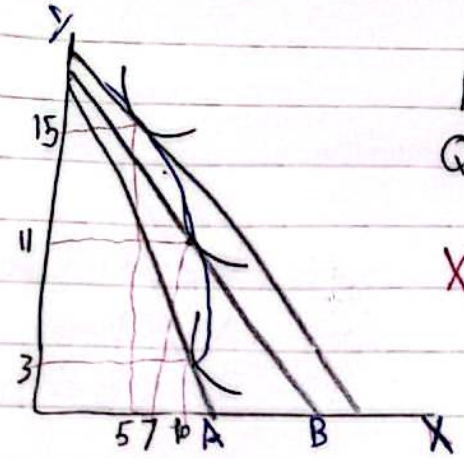
<1 inelastic D: $\% \Delta Q_D < \% \Delta P$, $Q_{D_X} \uparrow$ $Q_{D_X} \uparrow$

>1 elastic D: $\% \Delta Q_D > \% \Delta P$, $\% Q_{D_X} \uparrow$ $Q_{D_Y} \downarrow$

② downward sloping (PPC)



③



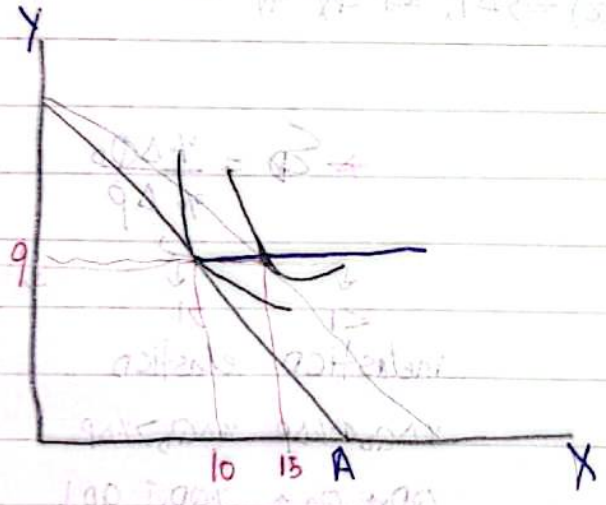
$P_x \downarrow$

$Q_x \downarrow$ $Q_Y \uparrow$

X ⟶ giffen good

④ unitary elastic

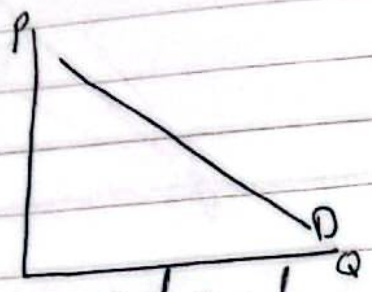
$\% \Delta P = \% \Delta QD$



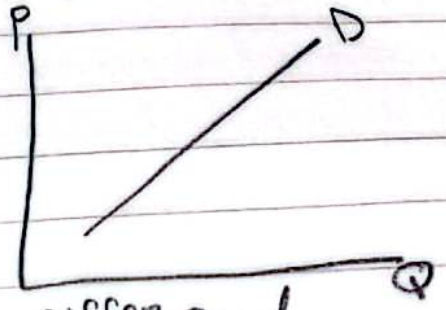
ε_x	Q_x	Q_Y	PCC
inelastic $\% \Delta Q_D < \% \Delta P$	↑	↑	up ward sloping
Elastic $\% \Delta QD > \% \Delta P$	↑	↓	downward sloping
Giffen good $P \downarrow$ $Q \downarrow$	↓	↑	back ward sloping
unitary elastic $\% \Delta QD = \% \Delta P$	↑	—	horizontal

Show graphically

① demand curve of a normal good and a giffen good

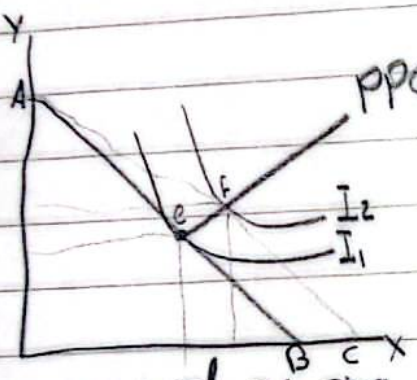


normal good

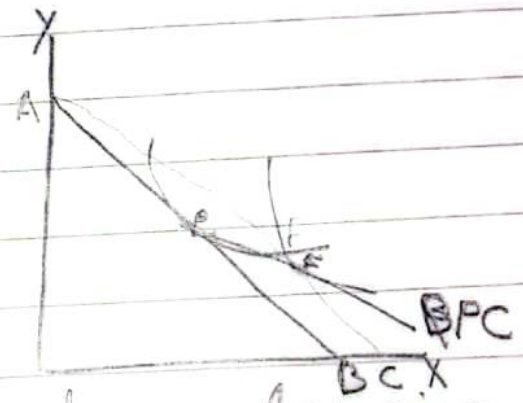


giffen good

② upward and downward sloping PPC

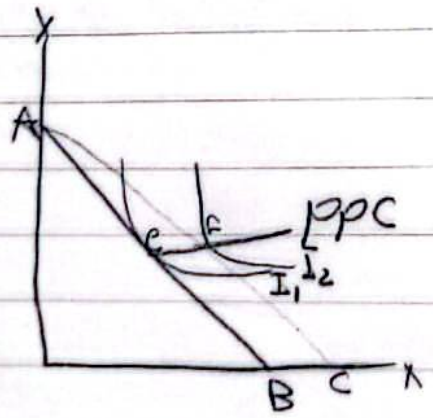


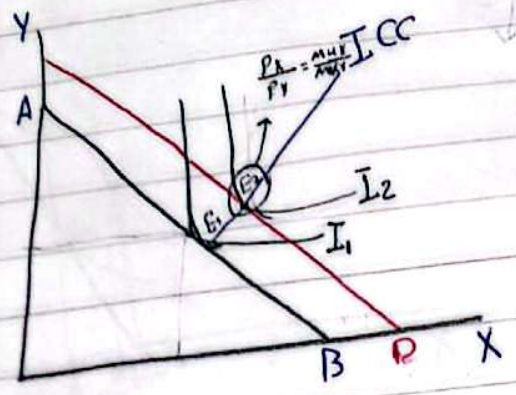
upward sloping



downward sloping

③ PPC incase of unitary elastic x.

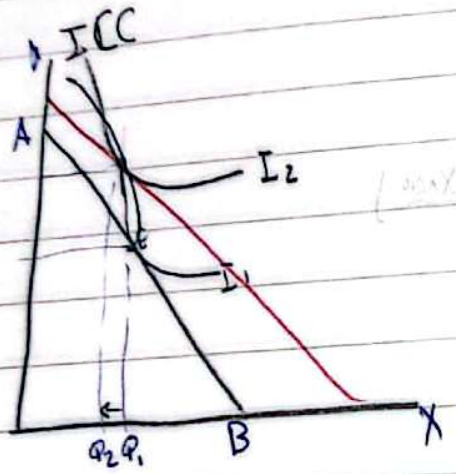




income ↑
X ↑
Y ↑

« X, Y → Normal goods »

« X → inferior good »



ICC is backward bending.

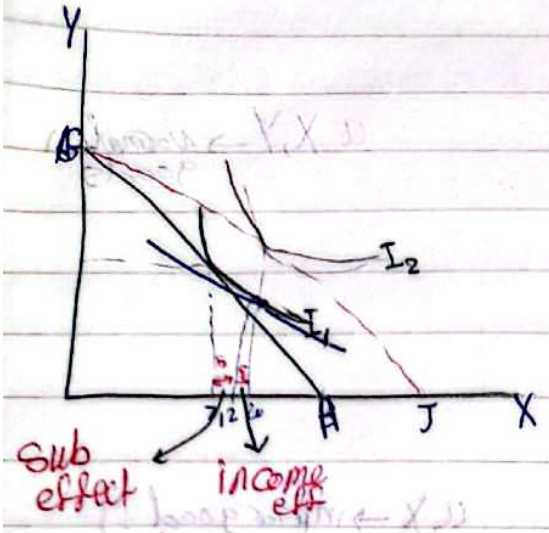
* Price effect = sub effect + income effect

price effect
$P_1 \Delta$

substitutional effect
P_X ↓
X ↑
Y ↓

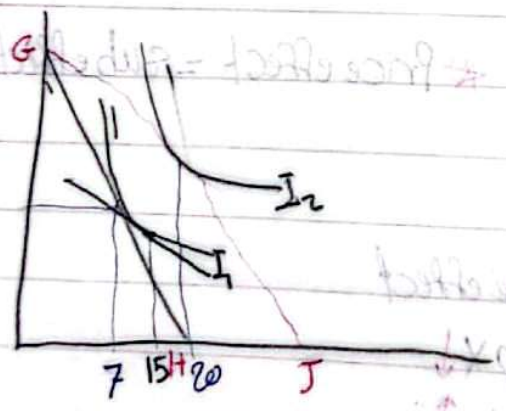
income effect
P_X ↓
X ↑
Y ↑

income > sub ⇒ complements
sub = 0 ⇒ perfect complements
Price effect = income effect

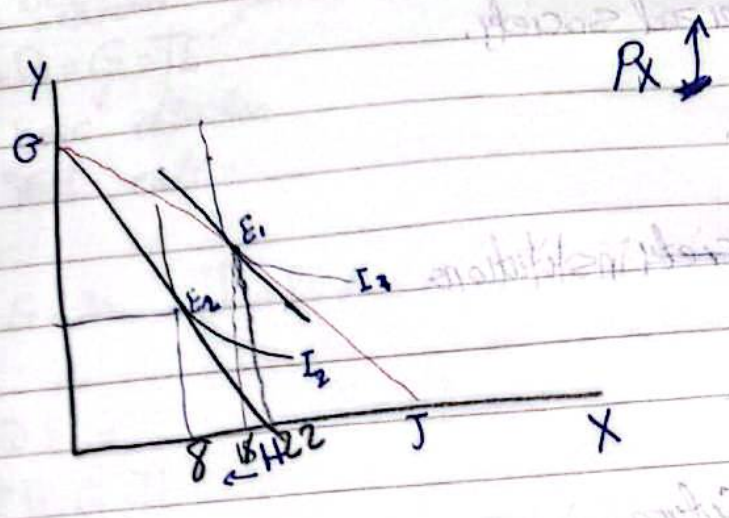


$P_X \downarrow$

* divide the price effect to income and sub.
  - draw new budget line
    - parrarel to new budget line (slope = $\frac{P_{X_{new}}}{P_Y}$)
    - tangent to the old indiff

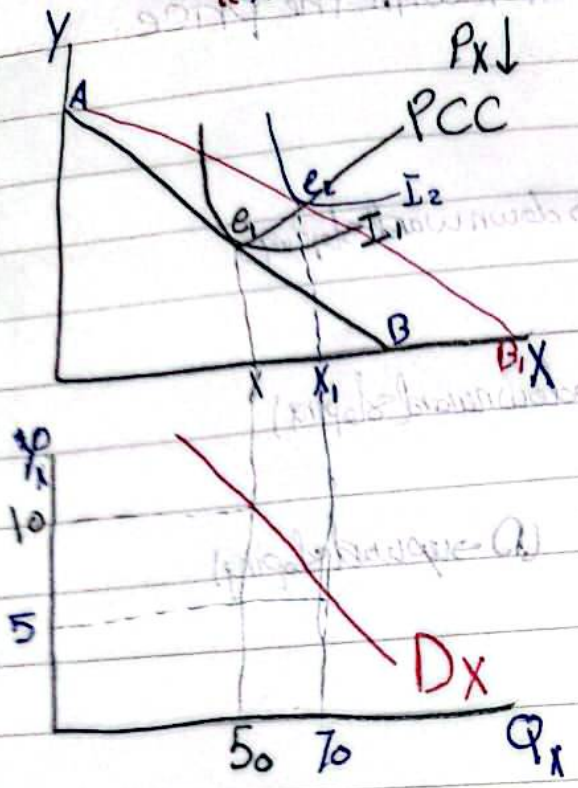


income effect = 5
sub effect = 8
price effect = 13



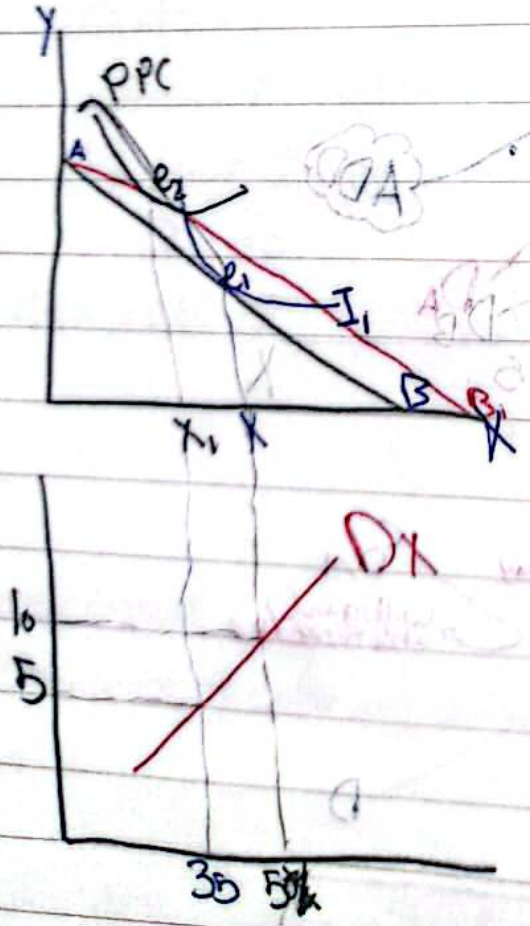
income eff = 16 - 8 = 8
sub effect = 22 - 16 = 6
price effect = 14

«X is normal good»



Px	X
10	50
5	70

«X is giffen good»



Px	X
10	50
5	35

Type	sub effect (Px↓)	income effect
Normal	increas	Increase Dx
Inferior	increase >	decrease
giffen	increase <	decrease.

→ مش مهم مين اطبر (D → downward sloping)

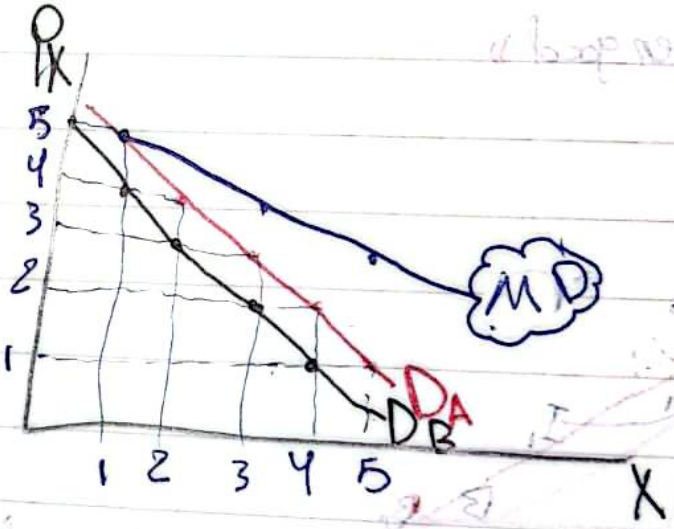
(D → downward sloping)

لازم اعرف مين اطبر
عشان اعرف هي انهي
نوع

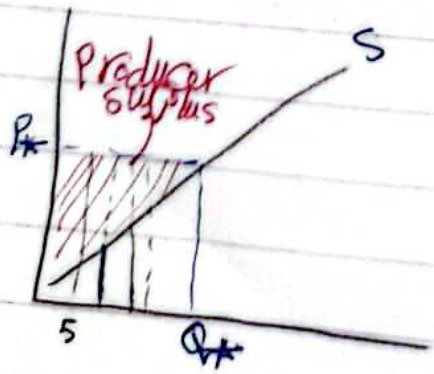
(D → upward sloping)

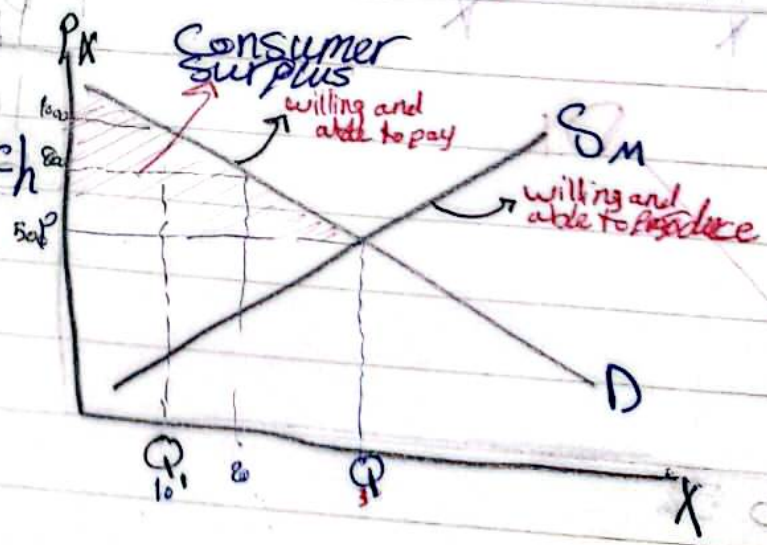
ex

P	D_A	D_B	
1	5	4	9
2	4	3	7
3	3	2	5
4	2	1	3
5	1	0	0

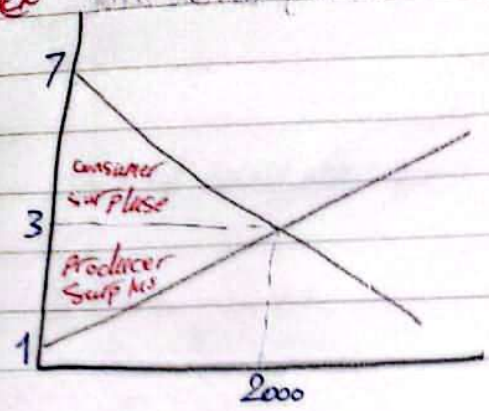


market is the horizontal sum of each demand





ex



- Consume surplus $= \frac{1}{2} \times 2000 \times 4 = 4000$
- producer surplus $= \frac{1}{2} \times 2000 \times 2 = 2000$
- total surplus $= 2000 + 4000 = 6000$

ex

$D: P_D = 12 - 0,5 Q_D$

$S: P_S = 0,1 Q_S$

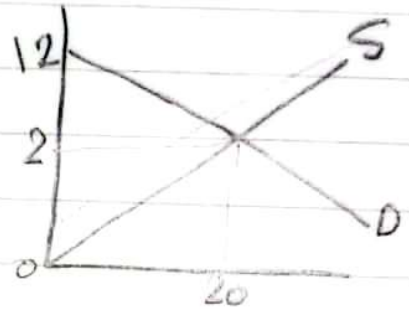
$P_D = P_S$

$12 - 0,5Q = 0,1Q$

$12 = 0,6Q$

$Q = 20$

$P = 0,1 \times 20 = 2$



consume surplus $= \frac{1}{2} \times 20 \times 10 = 100$.

producer surplus $= \frac{1}{2} \times 20 \times 2 = 20$.

total surplus $= 120$.

assignment

- Show graphically derivation of demand curve of Normal good.
  giffen good.
- difference between effect of increasing income on demand of normal good and inferior good
- $D: P_d = 10 - 2Q_d$
  $S: P_s = 1 + 0,5 Q_s$ find: consume, producer, total surplus.